Zimu Wang

zm.wang@berkeley.edu | 510-603-9827 | zimuwang-real.github.io | linkedin.com/in/zimu-wang-32b62a289

Education

University of California, Berkeley, Berkeley, CA — Graduation: May 2027

- Bachelor of Arts in Computer Science, GPA: 3.92/4.0
- Coursework: Neural Networks, Deep Reinforcement Learning, Computer Vision, Optimization, Probability Theory, Databases, Data Structures & Algorithms

Experience

ByteDance, LLM Engineer Intern — Summer 2026

RSIHub: github.com/simple-agent-lab/RSIHub

- Led the development of RSIHub, an open-source **recursive self-improvement** framework that evolves agent prompts, skills, harnesses, and code.
- Architected a modular evolution engine that isolates mutable agent components from fixed evaluators and records each generated candidate in Git, preserving score integrity and commit-level reproducibility.
- Shipped a plugin operator SDK with automatic discovery, subprocess isolation, and declarative schemas, enabling new operators and strategies without core changes.
- Demonstrated best full-benchmark gains of **+13.5 pts** on Terminal-Bench 2 with MiniSWE (56.2% $\rightarrow$ 69.7%) and **+9.3 pts** on Tau3 Banking with Codex (24.7% $\rightarrow$ 34.0%) across four self-improvement strategies and all 186 tasks in the two benchmark suites.

Simple Long Horizon Agent: github.com/simple-agent-lab/simple-long-horizon-agent

- Shipped an on-demand skill runtime with package discovery, compact prompt menus, explicit skill selection, and a bundled reusable skill library.
- Unified four agent presets behind a composable `AgentSession`/toolset API, standardizing session and tool lifecycles while adding config-driven MCP support to evaluation runners.

Shanghai Jiao Tong University, Researcher — Summer 2025 – Spring 2026

EffiSkill

- Designed and built EffiSkill, a two-stage LLM framework for program optimization that first mines reusable Operator and Meta Skills from pairs of slow and optimized programs, then retrieves relevant skills and composes plans to optimize new programs automatically without execution feedback.
- Built the inference and evaluation stack and benchmarked all **623 EffiBench-X tasks** in both Python and C++, with optimization success rates **3.69 to 12.52 percentage points** above the strongest baseline.

CodeJudge

- Built and scaled a pipeline to train Qwen3-8B on 48K examples to assess code correctness, using **8×A100 GPUs**, **SFT and reinforcement learning**, custom rewards, and distributed verl/vLLM rollouts.
- Increased agreement with ground-truth labels by **15 percentage points** (54% $\rightarrow$ **69%**) in held-out evaluation, outperforming **Qwen3-32B (66%)** while using only **25%** as many parameters.

University of California, Berkeley, Data 8 (Foundations of Data Science) Tutor — Fall 2024

- Led weekly tutoring sessions, a discussion section, and office hours, teaching Python-based data analysis and visualization alongside statistical methods such as hypothesis testing, regression, and classification.

Publications & Manuscripts

- *Beneath the Surface: What We Overlook When We Talk About Agent Evolution* [pdf]
  Xiaobo Wang, **Zimu Wang**, Siyu Ye, Jiaqi Li, Min Tang, Jun Bai, Yuling Shi, Ruobing Wang, et al.
  Under review at ICLR 2027.
- *RSIHub: A Composable Framework for Agent-Evolution Methods* [pdf] [code]
  Zimu Wang, Xiaobo Wang, Siyu Ye, Jiaqi Li, Silin Chen, Yuling Shi, Ruobing Wang, et al.
  NeurIPS 2026 Meta-Agents Workshop.
- *EffiSkill: Agent Skill Based Automated Code Efficiency Optimization* [pdf]
  Zimu Wang, Yuling Shi, Mengfan Li, Zijun Liu, Jie M. Zhang, Chengcheng Wan, Xiaodong Gu.
  Under review at EACL 2027.

Skills

Languages: Python, C++, Java, SQL, Bash
ML/Agents: PyTorch, Hugging Face Transformers, verl, vLLM, SFT, GRPO, RLHF, MCP, agent evaluation
Infra/Tools: Docker, Weights & Biases, Git, Linux, NumPy, Pandas